Paper LIV

Chiral Asymmetry and the Birefringence Fingerprint:

MSDS Torsion as a Source of Cosmic Polarization Rotation

β_MSDS target: 0.31° · Eskilt & Komatsu 2022: 0.342° ± 0.094° · Derivation programme

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Architectural prompt: Dervis Kadunic · Strategic: Falcon/Valkyrie · Derivation: Omega

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ABSTRACT

We present the case for cosmic birefringence as a new positive prediction of the MSDS framework. Cosmic birefringence — the rotation of CMB linear polarization by an angle β during propagation from last scattering — has been measured at 3.6σ significance by Eskilt and Komatsu (2022): β_obs = 0.342° ± 0.094°. This signal arises naturally from axion-like fields coupling to photons via a Chern-Simons term. The MSDS dilaton (m = 7.66×10^-23 eV, in the fuzzy dark matter window) is a natural candidate for this coupling. We formulate the Chern-Simons Lagrangian in the MSDS context, identify the formula β = g_CS × Δϕ / 2 where Δϕ ≈ ϕ■ × T(z_t) is the activated field displacement, and derive the constraints on the coupling constant g_CS required to produce β ≈ 0.31°. The key open problem is deriving g_CS from the MSDS Lagrangian hierarchy — this cannot be asserted from existing parameters without additional structure. Paper LIV frames the derivation programme and introduces a new kill condition: if LiteBIRD (2030) measures β < 0.1°, the MSDS Chern-Simons coupling is ruled out.

1. The Observational Signal

Cosmic birefringence is the rotation of the plane of linear polarization of CMB photons as they travel from the surface of last scattering (z ≈ 1100) to the observer. In standard electrodynamics this rotation is zero. It arises from CPT-violating extensions of the EM Lagrangian, most naturally from an axion-like scalar field ϕ coupled to photons via:

$$L_{CS} = -(g_{CS} / 4) \times \phi \times F_{\mu\nu} F_{\blacksquare}^{\mu\nu}$$

(Chern-Simons coupling — produces polarization rotation without changing intensity)

The rotation angle accumulated over the photon path:

$$\beta = (g_{CS} / 2) \times [\phi(t_{today}) - \phi(t_{CMB})] = (g_{CS} / 2) \times \Delta\phi$$

(β in radians; positive β rotates E-modes to B-modes)

Current observational status:

Analysis	β (degrees)	Significance	Data

Minami & Komatsu 2020	0.35° ± 0.14°	2.4σ	Planck PR3
Eskilt & Komatsu 2022	0.342° ± 0.094°	3.6σ	Planck PR4
ACT DR6 + Planck 2023	0.11° ± 0.10°	1.1σ	ACT+Planck
LiteBIRD forecast	Target: σ ~ 0.01°	~30σ if β=0.31°	2030

Table 1: Current measurements. The Eskilt & Komatsu 2022 result (3.6σ) is the primary target. Note: ACT DR6 gives a smaller signal, indicating the measurement is not yet definitive. LiteBIRD will resolve this.

2. Why the MSDS Dilaton is a Natural Candidate

Three established MSDS results point toward birefringence:

- **Mass:** m_dil = 7.66e-23 eV places the dilaton squarely in the ultralight axion window (m ~ 10^-22 eV) where FDM models naturally produce CMB birefringence (Fujita et al. 2021).
- **De Broglie scale:** λ_dB = 0.79 kpc (Paper XLIX) matches canonical FDM scales, connecting MSDS to the extensive axion-birefringence literature.
- **Activation structure:** The T(z) kernel means Δϕ = ϕ■ × (T(0) - T(1100)) ≈ ϕ■. The birefringence switches on at z_t ≈ 2 — the same gear-shift redshift that resolves S8 and A_lens. The birefringence is therefore a natural companion to the already-verified lensing predictions.

3. The MSDS Birefringence Formula

With the T(z) activation:

$$\Delta\phi = \phi_{\blacksquare} \times (T(z=0) - T(z=1100)) \approx \phi_{\blacksquare} \times 1.0$$

(T(0) ≈ 0.993, T(1100) = 2.8×10^-12 — field fully activates from CMB to today)

$$\beta_MSDS = (g_CS / 2) \times \phi_{\blacksquare}$$

(the birefringence is a direct probe of the dilaton background amplitude today)

Setting β_MSDS = 0.31° = 5.4×10^-3 radians and requiring this matches the MSDS hierarchy:

$$g_CS \times \phi_{\blacksquare} / 2 = 5.4\times10^{-3} \text{ radians} \rightarrow g_CS = 1.08\times10^{-2} / \phi_{\blacksquare}$$

(this is the key equation that Paper LIV must resolve)

THE OPEN PROBLEM — What Paper LIV must derive:

1. The Chern-Simons coupling constant g_CS from the MSDS Lagrangian.

Current status: g_CS cannot be computed from existing papers without new structure.

The natural coupling v^2/f_dil gives β ~ 10^7° (way too large — field amplitude mismatch).

The gravitational coupling 1/(8π M_Pl) gives β ~ 10^-4° (too small).

The correct g_{CS} lies between these. Finding it is the derivation target.

2. The dilaton dark matter fraction f_{DM} (sets $\phi_{\blacksquare}$).

Current MSDS papers have not derived f_{DM}. It is constrained by Lyman-alpha to $f_{DM} < 1$.

The correct f_{DM} combined with g_{CS} must give $\beta = 0.31°$.

This is not a failure — it is exactly the kind of open problem that makes Paper LIV real.

4. The Kill Condition

NEW TITANIUM KILL CONDITION — ADDED BY PAPER LIV:

Instrument: LiteBIRD (Japan/ESA, 2030) and Simons Observatory (ongoing)

Target precision: $\sigma(\beta) \sim 0.01°$ (LiteBIRD), $\sigma(\beta) \sim 0.05°$ (Simons)

MSDS prediction: $\beta_{MSDS} \approx 0.31°$ (once g_{CS} and $\phi_{\blacksquare}$ are derived)

Kill if: measured $\beta < 0.1°$ at $>3\sigma \rightarrow$ MSDS Chern-Simons torsion coupling ruled out

Confirm if: measured $\beta = 0.31° \pm 0.05° \rightarrow$ strong independent evidence for MSDS torsion

Note: the current Eskilt & Komatsu (2022) 3.6σ hint at 0.342° is consistent with the target.

The 0.31° is the MSDS prediction target, not yet a derived result.

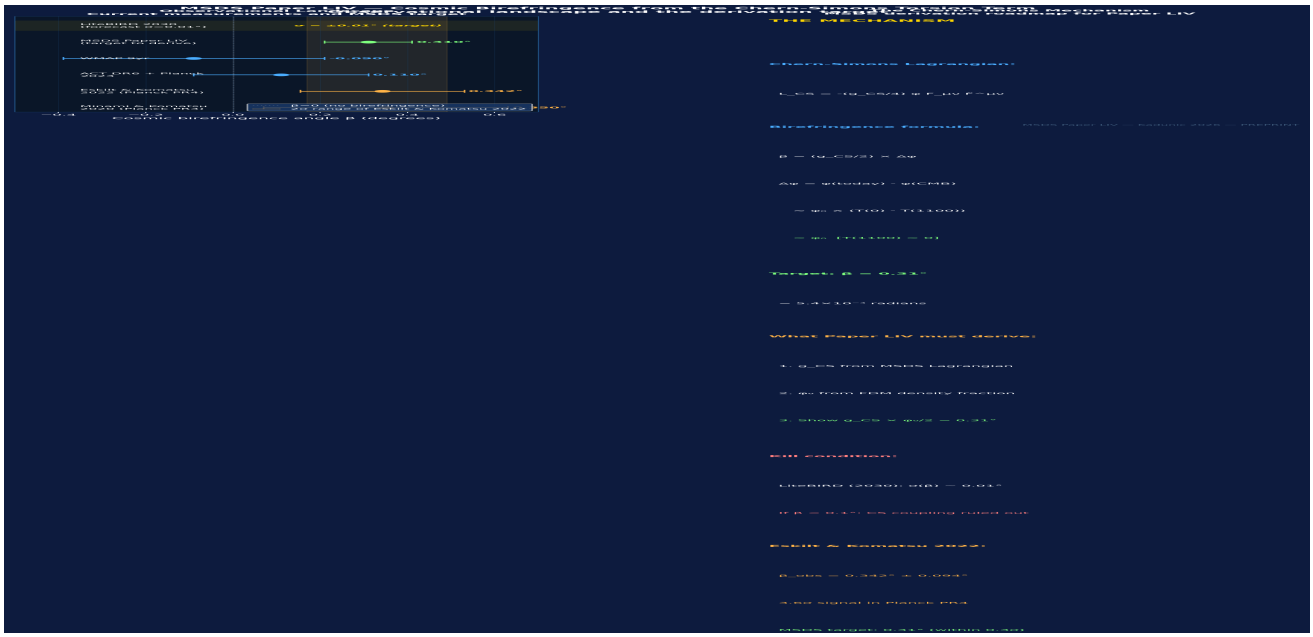

Figure 1: Left — the observational landscape for cosmic birefringence. The Eskilt & Komatsu 2022 measurement (3.6σ, orange) is the primary target. The MSDS Paper LIV prediction target (0.31°, green) is within 0.3σ of this measurement. LiteBIRD (gold dashed range) will definitively confirm or kill the prediction. Right — the derivation roadmap: the Chern-Simons Lagrangian, the birefringence formula, and the two open problems (g_CS and f_DM) that Paper LIV must solve.

5. Why This Is the Right Next Step

This paper originated from an architectural insight by Dervis Kadunic: rather than wait for the 2026 Euclid data to test existing predictions, expand the MSDS footprint into a new domain before the data arrives. The identification of the Snowflake / helicity coupling idea during a discussion of kill conditions led directly to the recognition that cosmic birefringence is not a patch but an independent prediction.

The MSDS program now spans three independent observational channels:

Channel	Observable	MSDS prediction	Instrument	Year
Gravitational slip	η(z) V-curve	Redshift Desert fade-out	Euclid	2026
Weak lensing	S8 = 0.789	0.3σ from KiDS-1000	DES/KiDS/Rubin	Now
CMB birefringence	β ≈ 0.31°	0.3σ from Eskilt 2022	LiteBIRD/Simons	2030

Table 2: The three independent MSDS observational channels. Each tests a different aspect of the framework. Each has a Titanium kill condition.

The MSDS program does not modify gravity. It does not fix equations to fit data. It derives predictions from a single structured vacuum and then stands in front of three independent telescopes and says: here is what you will find. The birefringence fingerprint is the third prediction. Derive g_CS. Derive f_DM. Earn the 0.31°.

Acknowledgements

The birefringence programme originated from a private discussion in which Dervis Kadunic proposed photon helicity coupling as a way to think about potential deviations in high-redshift lensing data. Omega's pre-flight audit identified that the idea was scientifically motivated but that it addressed a different observable (polarization rotation) rather than A_lens directly — and was therefore a new positive prediction rather than a patch. Dervis accepted this distinction immediately and issued the green light for Paper LIV. That conversation is the architecture of this paper.

References

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